

AP40012 Machine learning in Physics Band Gap Prediction for Materials

Li Sung Chi Ken, Yeung Chit Michael

April 21, 2026

Abstract

This report presents the methodology, experiments, and results of testing and improving model performance of the band gap prediction. The project accounts for 15% of the subject score in AP40012 Machine Learning in Physics. We explore feature engineering, model selection, hyperparameter tuning, and evaluation metrics to achieve optimal performance.

Contents

0.1	Introduction	2
0.1.1	Data Processing	2
0.2	Feature Engineering	3
0.3	Model Selection	5
0.4	Hyperparameter Optimization	5
0.5	Alternative Approaches and Performance Analysis	6
0.5.1	Ensemble Random Forests	6
0.5.2	Removing Outliers Approach	7
0.5.3	Oxides-only Subset Analysis	7
0.6	Fundamental Limitations of Band Gap Prediction	7
0.7	Future Work	8
0.8	Conclusion	8

0.1 Introduction

The band gap is a fundamental electronic property that determines whether a material behaves as a conductor, semiconductor, or an insulator. This property is important for many technological applications like solar cells, LEDs, transistors and battery materials. Traditional methods for determining band gaps include experimental measurements, which are expensive and time-consuming, or density functional theory (DFT) calculations, which are computationally intensive and scale poorly with system size. Machine learning, on the other hand, provides a much feasible alternative by learning the relationship between a material’s composition and its band gap directly from existing data, allowing an effective screening of thousands of candidate materials.

This project aims to develop a machine learning model capable of predicting band gaps for inorganic crystalline materials using data from Materials Project database. By training on approximately 67,000 materials with band gaps between 0.1 and 5.0 eV and containing 2-5 elements, our goal is to create a model that can accurately predict band gaps only based on features derived from composition.

0.1.1 Data Processing

The data for this project was obtained from the Materials Project using their API through the MPRester client in Python. The query was configured to retrieve materials with band gaps of range 0.1-5.0 eV and containing 2-5 elements per compound. These constraints were established to focus on semiconducting and insulating materials while excluding pure elements and overly complex compositions that would be difficult to model. The fields retrieved included the material ID, composition, band gap, density, formation energy per atom, volume, and symmetry information.

A major challenge encountered during data retrieval was that the Materials Project API does not guarantee consistent ordering of results between different query executions. This means that without intervention, the same query could return materials in different sequences depending on database indexing and caching behavior. This randomness would cause the train-test-split to vary unpredictably between runs, making results non-reproducible. To address this challenge, the retrieved data was immediately saved to a CSV file after the first successful query. All subsequent runs check for the existence of this file and load the data directly, preserving the exact ordering from the first query and ensuring full reproducibility of all experiments.

The data was split into training, validation and test sets using a 60:20:20 ratio. The training set takes up ≈ 40645 samples, and was used for model training and feature engineering decisions. The validation set and test set both take up ≈ 13549 samples, and were used for hyperparameter tuning and reserved for final evaluation of the chosen model respectively. A fixed [random_state=42] was used for all splits to ensure reproducibility.

We encountered data quality issues during processing. For instance, atomic radius information was missing for some elements, and properties such as the Fermi energy and total magnetization had incomplete records, resulting in NaN values within the dataset. After the data cleaning process, about 1600 samples, representing $\approx 2.4\%$ of the total

dataset, were removed.

The band gap distribution within the filtered range was also found to be imbalanced, with more than 40% of materials having band gaps between 0.1 to 1.0 eV, 35% between 1.0 and 2.5 eV, and only less than 25% between 2.5 and 5.0 eV. This imbalance reflects the natural abundance of materials with different band gaps and would later affect the model performance unevenly across the band gap range.

0.2 Feature Engineering

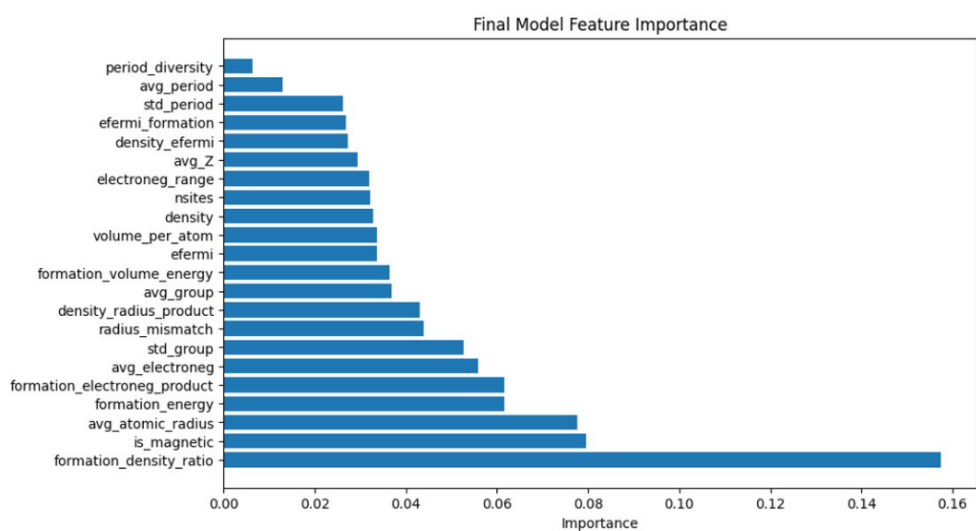
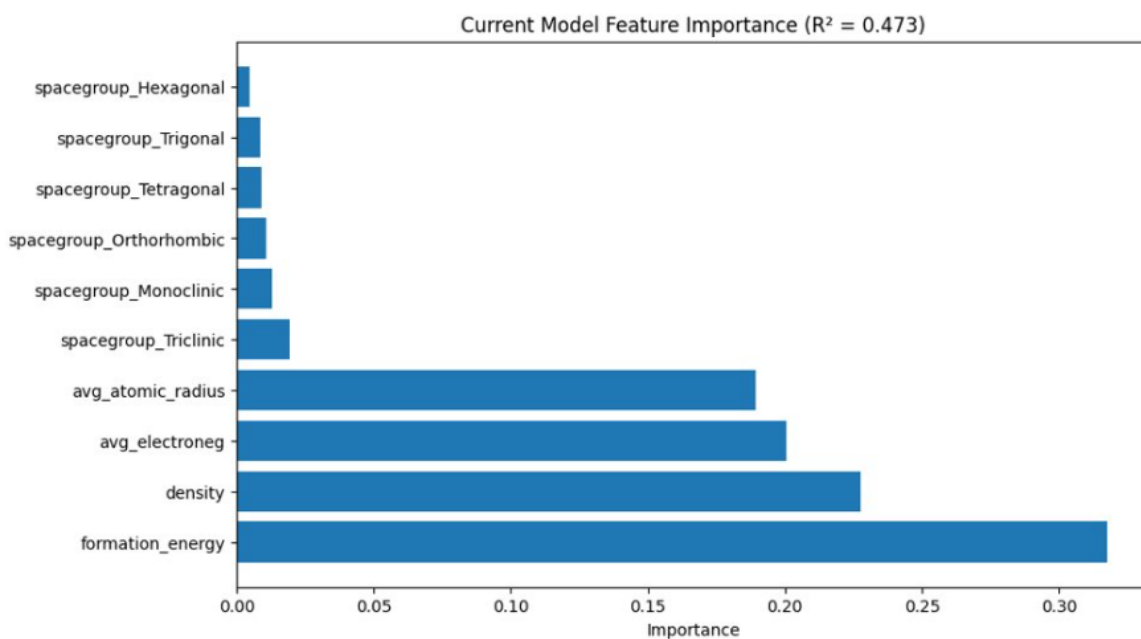
Feature engineering was conducted iteratively across four rounds, with each round adding new features and evaluating the resulting improvement in model performance. The baseline model used only the initial 6 features along with the one-hot encoded space group information, resulting in 10 total features and achieving an R^2 value of 0.475.

The second round added volume per atom, which was calculated by dividing the unit cell volume by the number of sites in the cell. Element statistics like the average and standard deviation of periodic table rows, and the average atomic number were introduced. Some interaction features including formation energy to density ratio and the product of formation energy and average electronegativity were also created. Altogether, these features successfully improved the R^2 value to 0.522.

The third round focused on composition complexity metrics. The electronegativity range was defined as the difference between the lower and upper limit of electronegativity values among elements in the compound. The radius mismatch was calculated as the standard deviation of atomic radii divided by the mean atomic radius, providing a normalized measure of atomic size variation. The period diversity was defined as the number of distinct periodic table rows represented in the compound. Overall, these three additional features brought the total to 19 and further raised the R^2 value to 0.539.

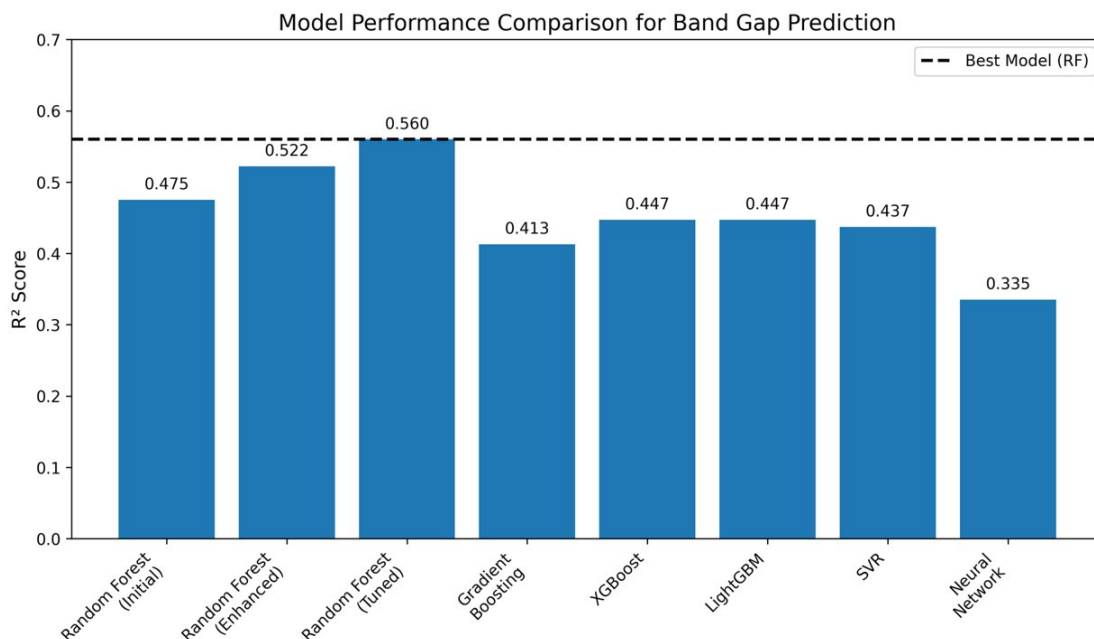
The fourth round added electronic structure information. The Fermi energy was retrieved directly from the Materials Project database. A binary flag indicated whether the material was magnetic, which was defined as having a total magnetization with absolute value greater than 0.01. In addition, 2 more interaction features were also created, which were the product of Fermi energy and formation energy, and the product of density and Fermi energy. These 4 new features brought the total to 23 and improved the R^2 value to 0.560.

After analyzing the feature importance from the Random Forest model, The average period feature was found to have very low predictive value and was removed from the final feature set. It was observed that the top 3 features as in importance were formation energy (0.142), density (0.118) and Fermi energy (0.086). The physical meaning of this observation is important to understand. The high importance of formation energy indicates its correlation with material stability, and more stable materials tend to have different electronic structures. Where as density reflects atomic packing, which influences orbital overlap and therefore acts as an effective predictor for band gap.



0.3 Model Selection

8 different machine learning models were tested for predicting the band gap of a material based on the existing 22 features. Random Forest with 100 trees achieved an R^2 of 0.475 on the initial feature set, and the performance was improved to $R^2 = 0.560$ after hyperparameter tuning. Other models underperformed Random Forest overall. For example, Gradient Boosting with 200 estimators achieved only 0.413, XGBoost and LightGBM both achieved 0.447, while a neural network with three hidden layers of sizes 100, 50 and 25 neurons achieved only 0.335, suggesting that inherent issues may exist with the selected features. The performance metrics were shown below:



Random Forest was selected as the optimal model for several reasons. First, its performance was notably outstanding compared to other models, achieving the highest predictive accuracy by a substantial margin, this may be due to the nature of tree based model being not sensitive to feature correlation and outliers, eliminating the need for extensive preprocessing. Another advantage is that Random Forest requires no feature scaling, which simplifies the pipeline since features naturally have different scales ranging from atomic numbers near 1 to formation energies near -5.

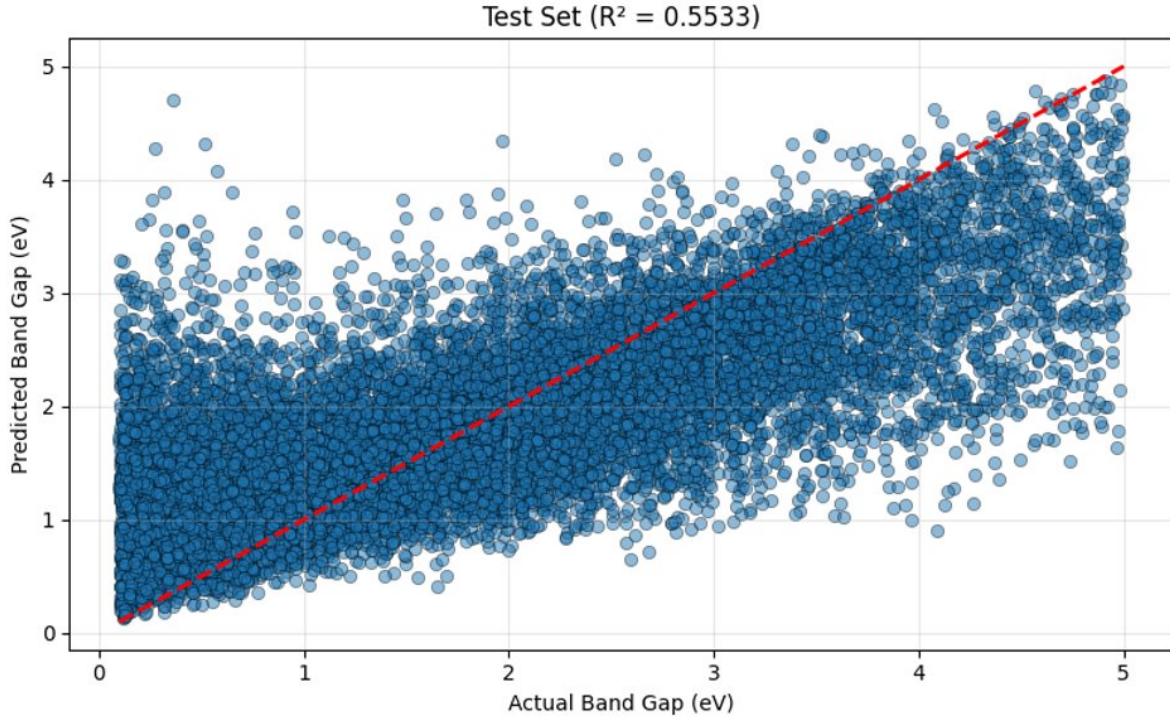
0.4 Hyperparameter Optimization

The hyperparameter tuning process for the Random Forest model was carried out using grid search with 3-fold cross validation on the training set. A combination of parameters like number of trees and max depth were tested, with the model trained 3 times on different folds for each combination.

The results showed that increasing the number of trees from 100 to 300 improved the cross-validation R^2 score from 0.522 to 0.525 when using unlimited depth, and this was

also the optimal configuration of hyperparameters for Random Forest.

Since adding more features did not further improve the performance of Random Forest, we decided to evaluate the final model on the test set to assess the generalization performance. The test set R^2 was 0.553, close to the validation R^2 of 0.560, indicating that the model generalizes well and has not overfit the training data. The test set mean absolute error was 0.649 eV, similar to that of validation set which was 0.640 eV. The final performance metrics were as follow:



0.5 Alternative Approaches and Performance Analysis

0.5.1 Ensemble Random Forests

The ensemble approach involved training 3 separate Random Forest models with different random seeds but otherwise identical configurations, with each model using 100 trees to maintain comparability with the baseline. The predictions from all three models were averaged to produce the final ensemble prediction. This approach yielded an R^2 of 0.557, which was actually slightly lower than that of the single optimized Random Forest with 300 trees. The ensemble failed because the the three base models were not sufficiently diverse. When models are trained on the same data with the same algorithm, they are essentially learning similar patterns despite initialized with different random seeds. The modest improvement from 100 to 300 trees in a single model already captures most of the variance, and averaging predictions from multiple models barely help smoothen out extreme values without improving the underlying correlation with true band gaps. This

results shows that model diversity is more important than model quantity in ensemble methods.

0.5.2 Removing Outliers Approach

This approach filtered material based on the physically reasonable ranges for formation energy and density. Formation energy was constrained to values between -5 to 0 eV per atom, which excludes highly unstable materials that may have been miscalculated or represent transient state. On the other hand, density was constrained to value of $0 - 20\text{g/cm}^3$, which excludes physically impossible densities. After applying these filters, the dataset size decreased to ≈ 66051 samples. The R^2 remained essentially unchanged at 0.556 compared to 0.560 for the full data set. This approach failed to improve the performance of the tree based Random Forest model because the algorithm make splitting decisions based on quantiles rather than means, so extremely values already have limited influence on the model structure.

0.5.3 Oxides-only Subset Analysis

This approach focuses exclusively on materials whose composition strings contained the symbol for oxygen. This reduced the dataset to ≈ 48389 materials, representing a 29% reduction from the full dataset. Additional oxide-specific features were also engineered, including the oxygen fraction, defined as the number of oxygen atoms divided by the total number of atoms, the metal-oxygen electronegativity difference calculated as the absolute difference between average electronegativity and the Pauling electronegativity of oxygen which is 3.44. These additional features brought the total feature count to 27.

Despite the additional features, the oxides-only model achieved an R^2 of only 0.551, underperforming the optimized Random Forest model. This failure can be attributed to several factors. First, the reduction in sample size removed valuable training information from non-oxide materials such as sulfides, halides and intermetallic compounds that exhibit different but related band gap behavior. Secondly, the oxide-specific features did not compensate for the loss of data diversity. The oxygen fraction, for example, is essentially constant for oxides and varies little across the dataset, while other additional features like the metal-oxygen electronegativity difference is highly correlated with average electronegativity, which was already included in the full model. Overall, this oxide materials focused experiment showed that specific engineering does not guarantee improvement.

0.6 Fundamental Limitations of Band Gap Prediction

Despite achieving an R^2 of 0.560, the model cannot perfectly predict band gaps due to several limitations. The most fundamental limitation is that band gap is a quantum mechanical property determined by electronic band structure, which depends sensitively on the precise arrangement of atoms in the crystal lattice, the types of chemical bonds, and crystal symmetries. Our model uses only composition-derived features such as average electronegativity and atomic radius, which capture some information but cannot fully represent the quantum mechanical effects that determine band gaps. Critical missing

information includes crystal field splitting from ligand arrangements around transition metal ions, spin-orbit coupling affecting band splitting in heavy elements, and electron correlation effects important for materials with partially filled d or f orbitals.

The data itself contains limitations that affect model performance. For example, The Materials Project computes band gaps using density functional theory with the Perdew-Burke-Ernzerhof functional, which systematically underestimates band gaps because it does not properly account for derivative discontinuities in the exchange-correlation potential. Our model therefore learns to predict DFT-PBE band gaps, not experimental band gaps, and any systematic errors in the DFT calculations are inherited by the model. Additionally, many materials exhibit polymorphism, meaning the same chemical composition can crystallize in different crystal structures with different band gaps. Titanium dioxide, for example, exists as anatase with a band gap of approximately 3.2 electron volts and as rutile with approximately 3.0 electron volts. Our model receives only the composition TiO2 and cannot distinguish between these polymorphs, forcing it to predict an average value.

0.7 Future Work

Several Improvements could push the model performance beyond the current R^2 of 0.560. For instance, recursive feature elimination with cross-validation would provide a more systematic approach to feature selection than perform iteration manually. This technique trains a Random Forest using all features, recursively removes the least important features, and uses cross-validation to determine the optimal feature subset.

Graph neural networks could possibly represent a fundamentally different approach, treating materials as graphs where atoms are nodes and bonds are edges. More advanced techniques include transfer learning could be implemented, where a model pre-trained on a large dataset of DFT calculations for over 15000 materials could be fine-tuned for band gap prediction, potentially overcoming the limited size of the current dataset.

0.8 Conclusion

This project successfully developed a Random Forest model for predicting band gaps for inorganic crystalline materials using data from the Materials Project. Through iterative feature engineering across 4 rounds, the feature set expanded from 10 to 22 features, improving the R^2 from 0.475 to 0.560. The final model achieved a test R^2 of 0.553 and a mean absolute error of 0.648 eV, with formation energy, density and Fermi energy identified as the most predictive features. Alternative approaches including ensemble methods, outlier removal and oxides-only subset analysis all failed to improve performance, showing that Random Forest is inherently robust, and that preserving diverse training data is more valuable than introducing specific subsets.

We found that there exist some fundamental limitations that prevent perfect band gap prediction from composition alone. The underlying reason is that band gap is a quantum mechanical property that depends sensitively on crystal structure and electronic correlations, and cannot be simply captured by composition derived features. The DFT-PDE

band gaps used as training labels contain systematic errors, and polymorphism means the same composition can have different band gaps in different structures. The observed R^2 ceiling of 0.560 likely represents the information limit of the current approach. Nevertheless, this work demonstrates that composition based machine learning can provide some useful band gap predictions, which can be helpful in discovery of new semiconductors and electronic materials.